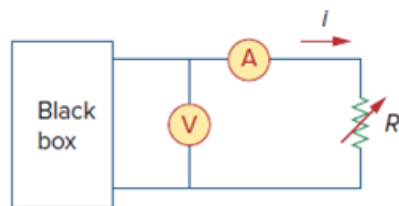


Problem

A black box with a circuit in it is connected to a variable resistor. An ideal ammeter (with zero resistance) and an ideal voltmeter (with infinite resistance) are used to measure current and voltage as shown in Fig. 4.143. The results are shown in the table on the next page.

Figure 4.143:



(a) Find i when $R = 12\ \Omega$.

(b) Determine the maximum power from the box.

$R(\Omega)$	$V(V)$	$i(A)$
2	6	3
8	16	2
14	21	1.5

Step-by-step solution

Step 1 of 9

Refer to Figure 4.143 in the text book.

Replace the Black box with the Thevenin's equivalent circuit including the load resistance circuit as shown in Figure 1.

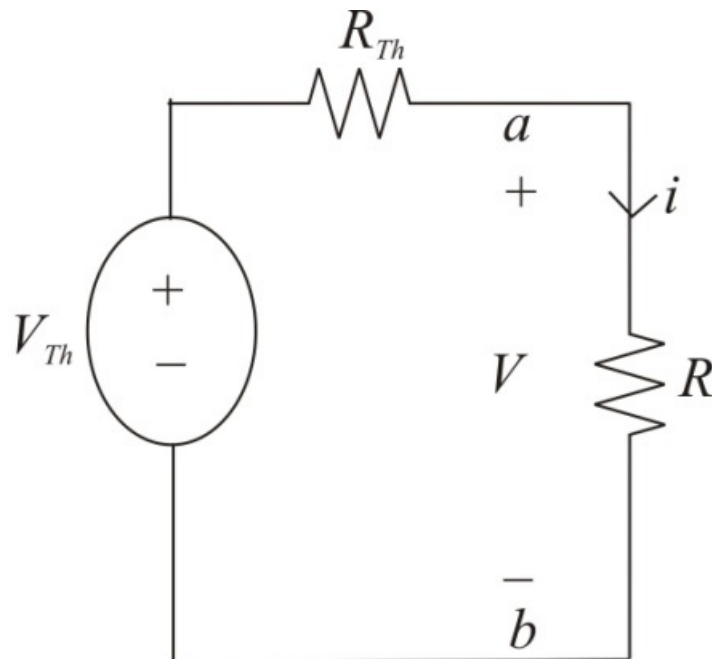


Figure 1

Step 2 of 9

Consider the expression for the Thevenin's voltage V_{Th} .

$$V_{Th} = iR_{Th} + V \dots\dots (1)$$

Step 3 of 9

Recall equation (1).

$$V_{Th} = iR_{Th} + V$$

Substitute 6 V for V and 3 A for i .

$$V_{Th} = 3R_{Th} + 6 \dots\dots (2)$$

Step 4 of 9

Recall equation (1).

$$V_{Th} = iR_{Th} + V$$

Substitute 16 V for V and 2 A for i .

$$V_{Th} = 2R_{Th} + 16$$

Substitute $3R_{Th} + 6$ for V_{Th} .

$$3R_{Th} + 6 = 2R_{Th} + 16$$

$$3R_{Th} - 2R_{Th} = 16 - 6$$

$$R_{Th} = 10 \text{ } \Omega$$

Therefore, the value of the Thevenin's resistance R_{Th} is, $10 \text{ } \Omega$.

Step 5 of 9

Recall equation (2).

$$V_{Th} = 3R_{Th} + 6$$

Substitute $10 \text{ } \Omega$ for R_{Th} .

$$V_{Th} = 3(10) + 6$$

$$= 30 + 6$$

$$= 36 \text{ V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, 36 V .

Step 6 of 9

(a)

Connect the load of $12\ \Omega$ to the Thevenin's circuit.

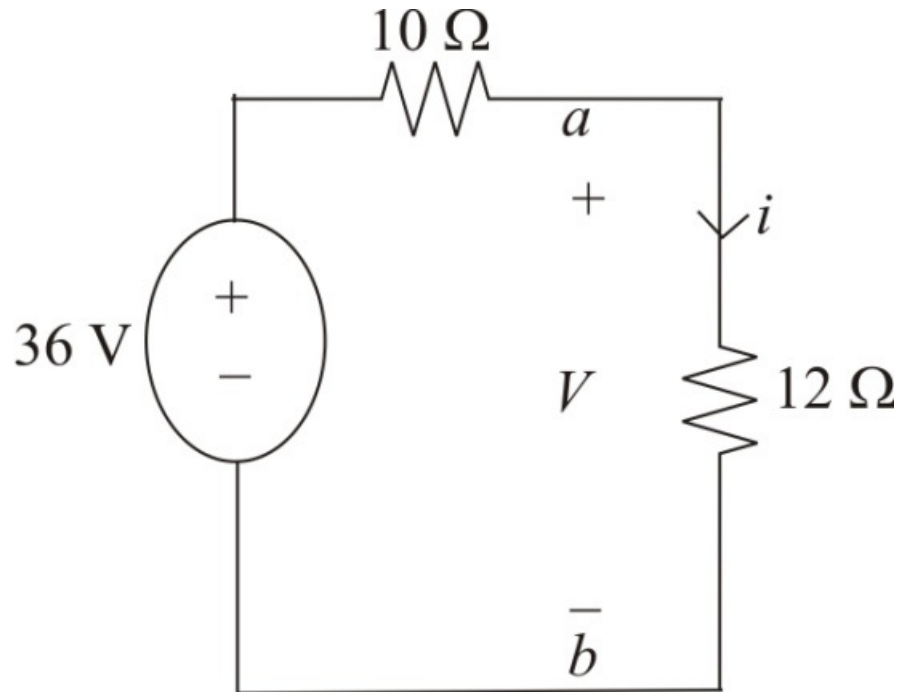


Figure 2

Step 7 of 9

Calculate the value of the current i .

$$\begin{aligned} i &= \frac{36}{10+12} \\ &= \frac{36}{22} \\ &= 1.636\text{ A} \end{aligned}$$

$$V_{ab} = 32\text{ V}$$

Therefore, the value of the current i is, 1.636 A .

[Comments \(1\)](#)



Anonymous

Vab is wrong. You should get $V_{ab} = IR = 1.636 \times 12 = 19.632\text{ V}$.

Step 8 of 9

(b)

For the maximum power to be delivered to the load, the value of the load must be equal to the Thevenin's resistance R_{th} . Therefore, the value of the load R is, $10\ \Omega$.

Step 9 of 9

Calculate the maximum power absorbed by the load.

$$P_{\max} = \frac{V_{th}^2}{4R}$$

Substitute 36 V for V_{th} and 10 Ω for R .

$$\begin{aligned} P_{\max} &= \frac{(36)^2}{4(10)} \\ &= \frac{1296}{40} \\ &= 32.4 \text{ W} \end{aligned}$$

Therefore, the value of the maximum power absorbed by the load is, 32.4 W.